

Partial evaporation in geothermal Rankine cycles: butane, cyclopentane and Novec 649

Partial evaporation in organic Rankine cycles can improve heat recovery from sources with a temperature glide by reducing the isothermal evaporation duty and modifying the temperature profile of heat addition. However, these benefits must compensate for any reduction in expander efficiency associated with two-phase operation. This study examines how the minimum permissible heat-source outlet temperature and expander efficiency determine when partial evaporation increases net power production. Simple cycles using n-butane and cyclopentane and a recuperated cycle using Novec 649 are optimized for a 140 °C water source and 20 °C cooling-water inlet. A sensitivity analysis comprising 360 optimized cases varies the minimum source outlet temperature from 135 to 20 °C and the expander efficiency from 60% to 90%. Each configuration is compared with its corresponding superheated cycle at 90% expander efficiency, using net power per unit source flow as the comparison basis. At equal expander efficiencies of 90% and a minimum source outlet temperature of 70 °C, partial evaporation increases net power by 6.5%, 15.2%, and 13.1% for n-butane, cyclopentane, and Novec 649, respectively. Relaxing this limit to 20 °C increases the gains to 40.8%, 59.3%, and 42.7%, through combined improvements in heat recovery and heat conversion. At this limit, gains remain positive with an 80%-efficient expander, reaching 22.8%, 40.6%, and 26.2%, respectively; only cyclopentane retains a positive gain at 60% efficiency. The corresponding designs at 90% efficiency require wet-to-wet expansion for n-butane and cyclopentane, whereas Novec 649 achieves wet-to-dry expansion. These results identify deeper source cooling as a major driver of the benefit of partial evaporation and quantify the expander performance needed to retain that benefit.

- [Partial evaporation in geothermal Rankine cycles: butane, cyclopentane and Novec 649](#)
 - [1. Boundary conditions and optimization formulation](#)
 - [2. Comparison of fluid and cycle configurations](#)
 - [3. Sensitivity rationale and initialization](#)
 - [4. Butane: simple cycle](#)
 - [5. Cyclopentane: simple cycle](#)
 - [6. Novec 649: recuperated cycle](#)
 - [7. Cross-fluid interpretation and limits](#)

1. Boundary conditions and optimization formulation

1.1 Common conditions

Parameter	Value
Source water	140 °C, 10 bar

Parameter	Value
Source minimum outlet	70 °C for the six-case comparison; 140–20 °C in 5 K steps for sensitivities
Cooling water	20 °C inlet; 24.99–25.00 °C outlet
Net cycle power used to size every design	1 MW after the working-fluid pump, before external-water pumps
Working fluids and architectures	Simple n-butane; simple cyclopentane; recuperated Novec 649
Expander efficiency	90% for superheated baselines; 90%, 80%, 70%, 60% for partial evaporation
Pump efficiencies	70%
Exchanger pressure losses	1% per active side
Minimum exchanger approach	5 K at 100 evaporator/condenser locations; 25 recuperator locations
Working-fluid pump inlet	At least 1 K subcooling
Superheated formulation	2 K inlet superheat and dry expander outlet
Partial-evaporation formulation	Optimized inlet state; no fixed quality or dry-outlet constraint

1.2 Optimization variables

Variable	Role
Source outlet temperature	Amount of source cooling, subject to the reinjection limit
Cooling-water outlet temperature	Approximately 5 K total warming
Working-fluid pump inlet pressure and enthalpy	Low-pressure state and subcooling
Expander inlet pressure and enthalpy	High-pressure state and inlet phase
Recuperator effectiveness	0–1, Novec only

Mass flows follow from the states, heat balances and 1 MW sizing target. Pressure and enthalpy bounds are broad and property-based. Each initial state comes from a saved coarse-grid optimum; the partial-evaporation formulation does not constrain the inlet quality.

1.3 Fixed-reference efficiency and comparison basis

The efficiency contours and baseline performance curves use net system power after all pumps divided by the source heat available between **140 and 20 °C**, at the source pressure of 10 bar:

$$\eta_{\text{fixed}} = \text{net system power} / [\text{source mass flow} \times (\text{hwater}(140 \text{ °C}, 10 \text{ bar}) - \text{hwater}(20 \text{ °C}, 10 \text{ bar}))]$$

The reference enthalpy drop is **504.723 kJ/kg** for every case. This efficiency is directly proportional to net system power per source flow, so deeper permissible cooling is assessed on the same energy basis. The 1 MW cycle sizing does not imply equal geothermal flow.

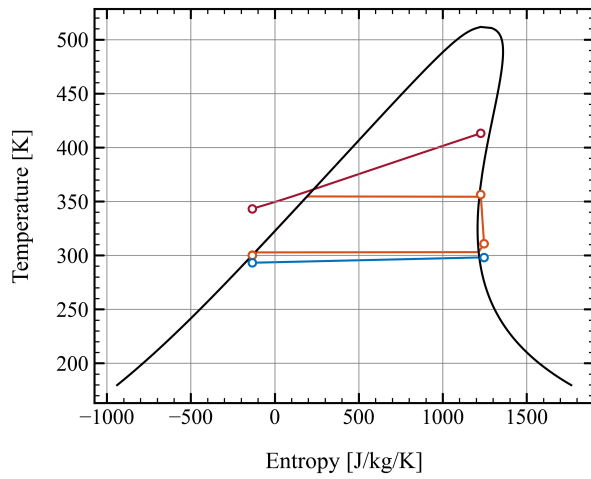
2. Comparison of fluid and cycle configurations

All six designs use a 70 °C reinjection limit and 90% expander efficiency. This table uses **superheated cyclopentane** as the common baseline; the sensitivities below use each fluid's own baseline.

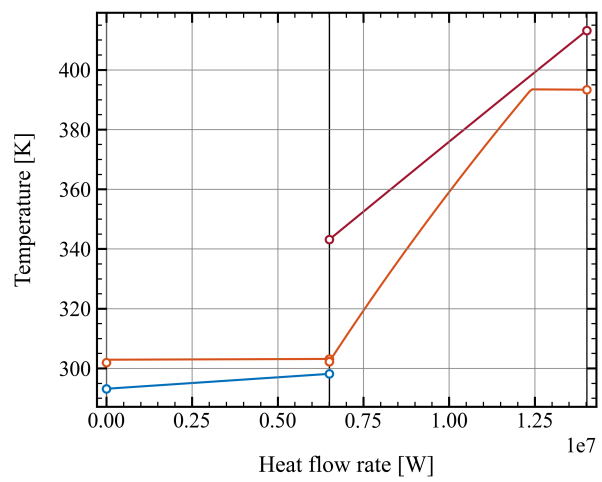
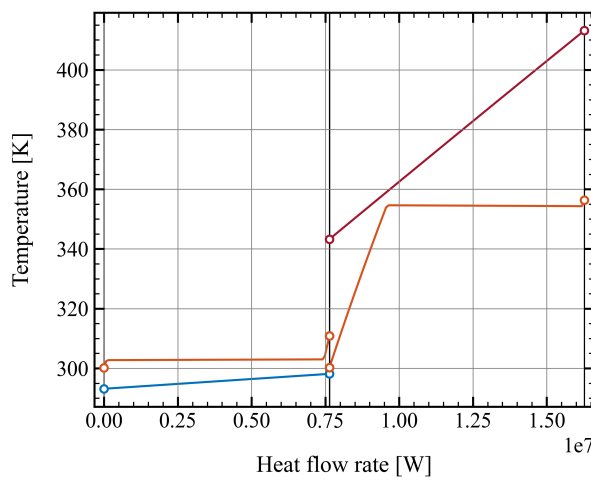
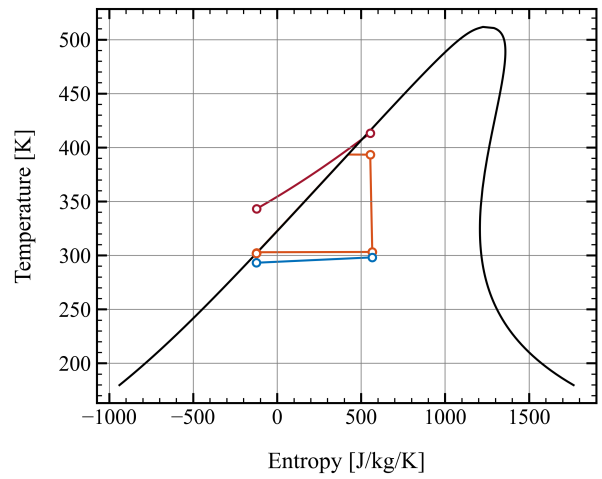
Case	Inlet vapor [%]	Outlet vapor [%]	Source outlet [°C]	Gain vs cyclopentane baseline [%]	Variable-reference system efficiency [%]	Fixed-reference system efficiency [%]
cyclopentane superheated	100.00	100.00	70.11	+0.00	11.5486	6.7663
cyclopentane partial evaporation	15.95	51.32	70.00	+15.24	13.3084	7.7974
butane superheated	100.00	100.00	70.48	+6.17	12.2615	7.1840
butane partial evaporation	57.00	85.40	70.00	+13.03	13.0528	7.6476
novec superheated	100.00	100.00	70.00	+8.92	12.5792	7.3702
novec partial evaporation	60.80	100.00	70.00	+23.20	14.2273	8.3358

Each fluid has a separate 2 × 2 comparison: superheated and partial-evaporation designs occupy the left and right columns, with T–s diagrams above T–Q diagrams.

Cyclopentane / simple
2 K superheat

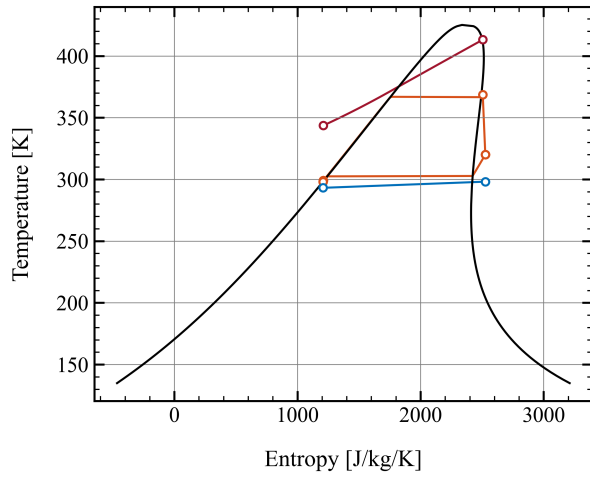


Cyclopentane / simple
Partial evaporation

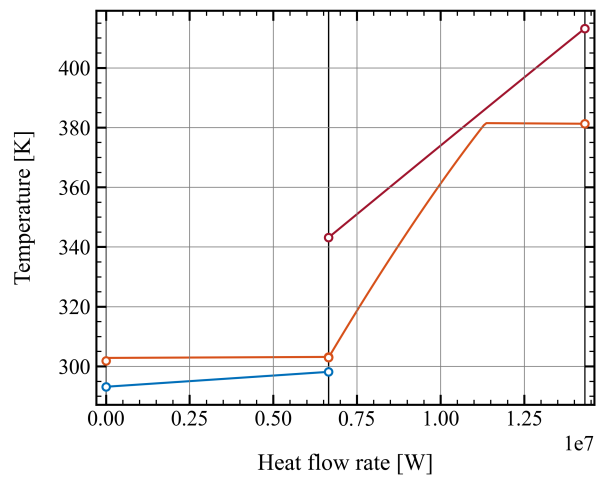
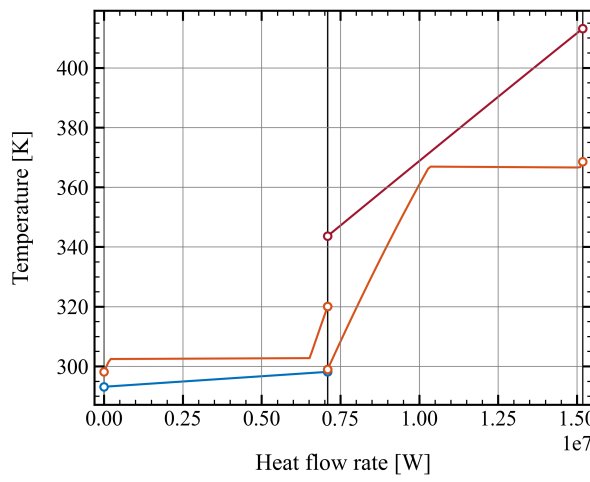
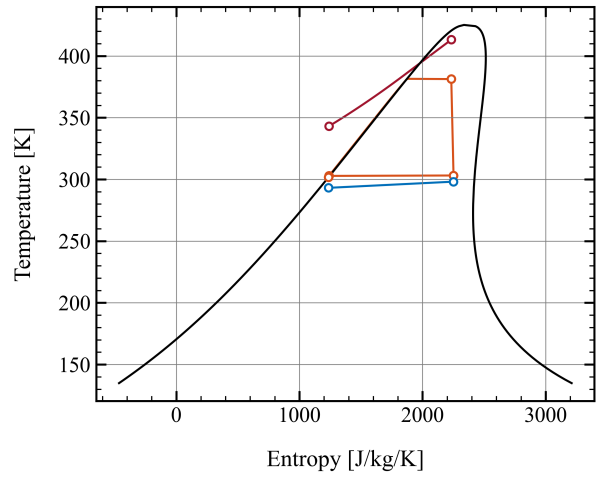


Working fluid Heat source Cooling water

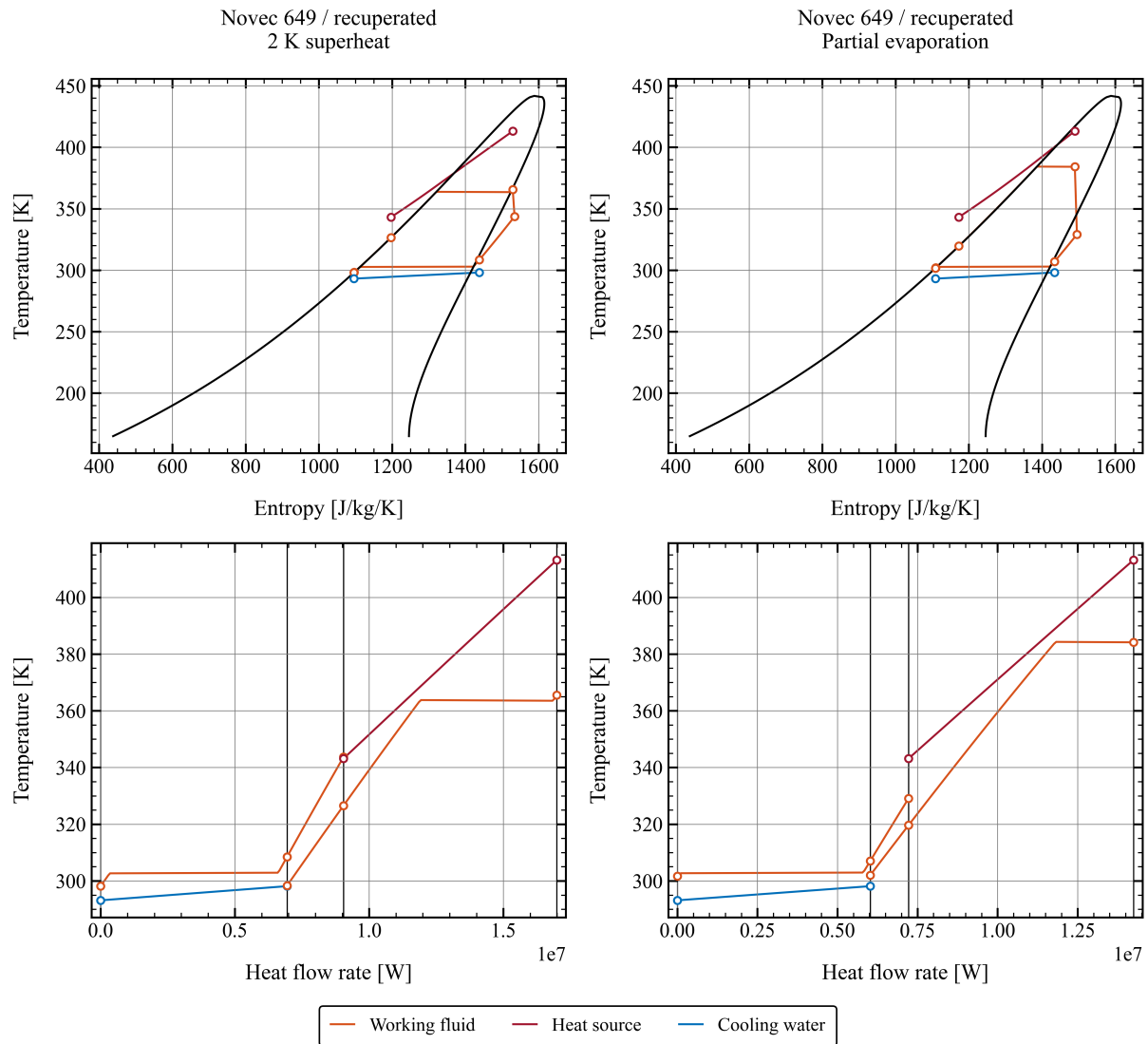
Butane / simple
2 K superheat



Butane / simple
Partial evaporation



Working fluid Heat source Cooling water



Relative to each fluid's own superheated design, the partial-evaporation gains at this limit are **+6.45% for Butane**, **+15.24% for Cyclopentane**, **+13.10% for Novec 649**. The highest fixed-reference efficiency is **8.34%**, for **novec partial evaporation**. The optimized source outlets differ, so the comparison includes both heat-recovery and heat-conversion effects.

3. Sensitivity rationale and initialization

3.1 Physical question and grid

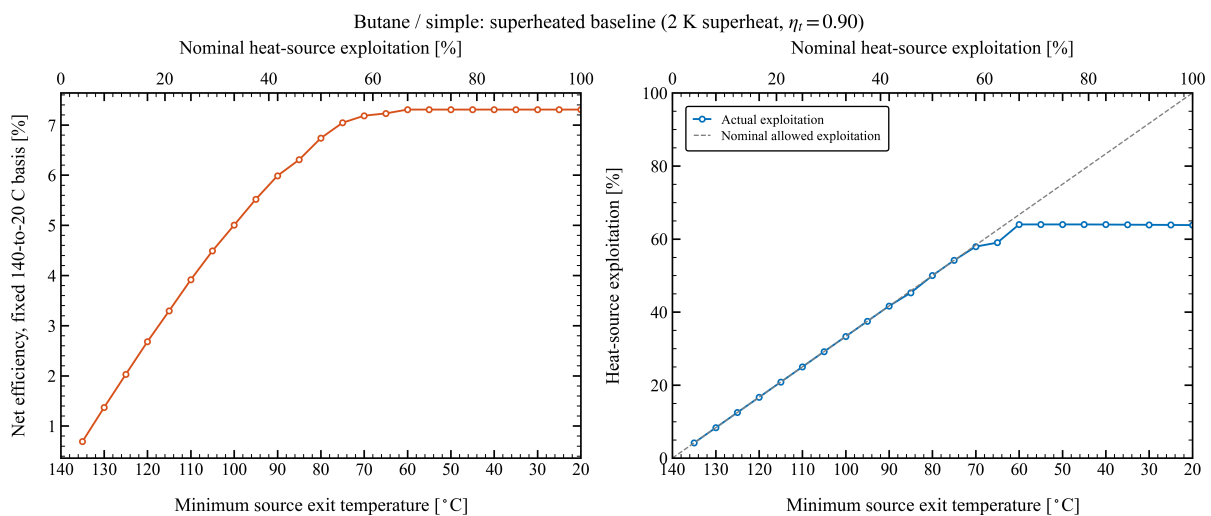
Wet admission can reduce the isothermal evaporation duty and alter source matching, while an efficiency penalty reduces work recovered in expansion. The sensitivity tests where the former benefit outweighs the latter. There are **24 finite-heat temperatures × five designs = 120 optimized points per fluid**, or **360 total**. The 140 °C limit permits no geothermal temperature drop and makes ThermoOpt's available-heat denominator zero. Its 15 requested combinations are recorded as unavailable, not as zero-efficiency cycles.

3.2 Plot conventions

Nominal exploitation is $100 \times (140 - \text{minimum outlet}) / 120$; actual exploitation replaces the minimum with the optimized outlet. Both use 20 °C as the reference. Quality is shown in percent: 0% includes liquid and 100% includes superheated vapor. Dark blue indicates more liquid. Contour lines label the interpolated bands. The T-s/T-Q overlays use every other temperature at 90% expander efficiency, thin lines, and magma restricted to colormap coordinates 0.25–0.75. Their native cumulative heat-duty coordinate uses the 1 MW sizing basis.

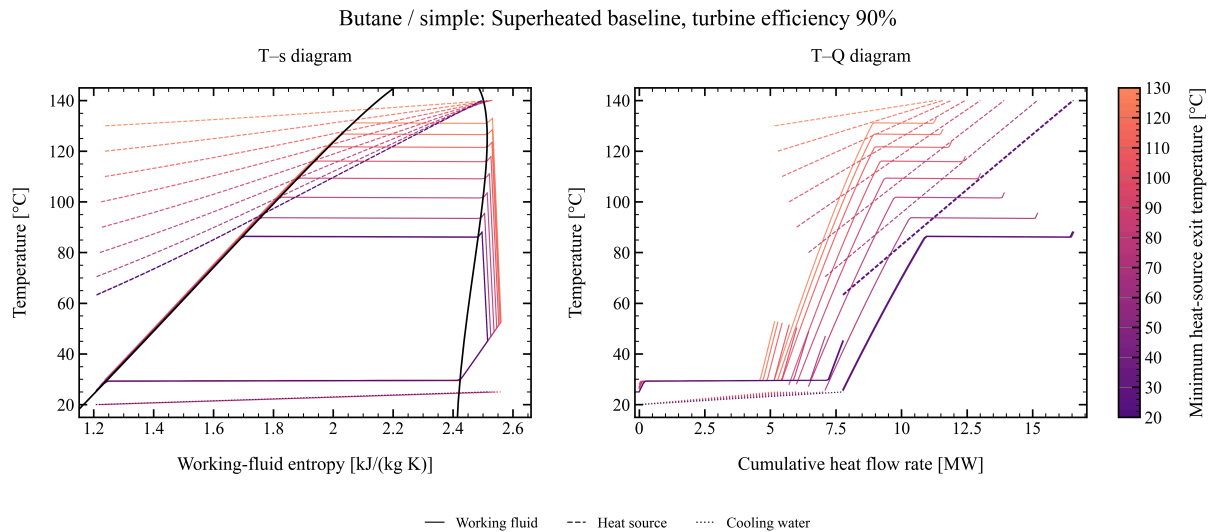
4. Butane: simple cycle

4.1 Superheated baseline



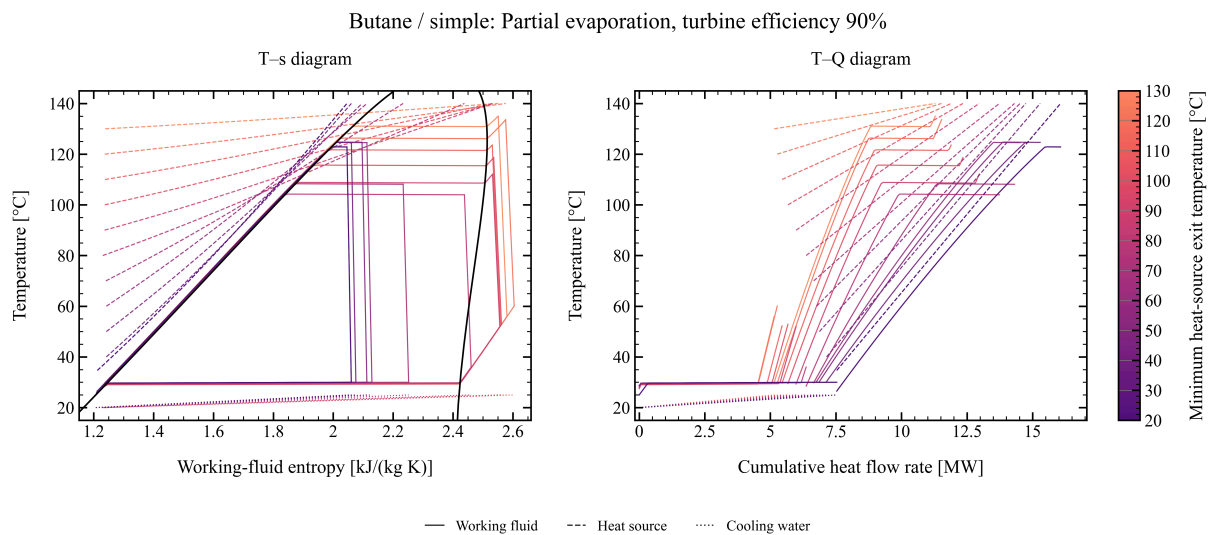
At 90% expander efficiency, the baseline's fixed-reference net efficiency changes from **0.69%** at the 135 °C reinjection limit to **7.31%** at 20 °C. Its corresponding net power per unit source flow changes from **3.48 to 36.88 kW per kg/s**. These two measures have exactly the same trend because their heat reference is fixed.

At the 20 °C limit, the baseline actually leaves the source at **63.41 °C**, or **63.82% actual exploitation**. Permission for further cooling does not require the optimizer to use it: the selected pressure levels, phase-change temperatures and exchanger approaches determine useful recovery.



The overlays show how the pressure levels and heat-addition profile change as the reinjection limit is relaxed.

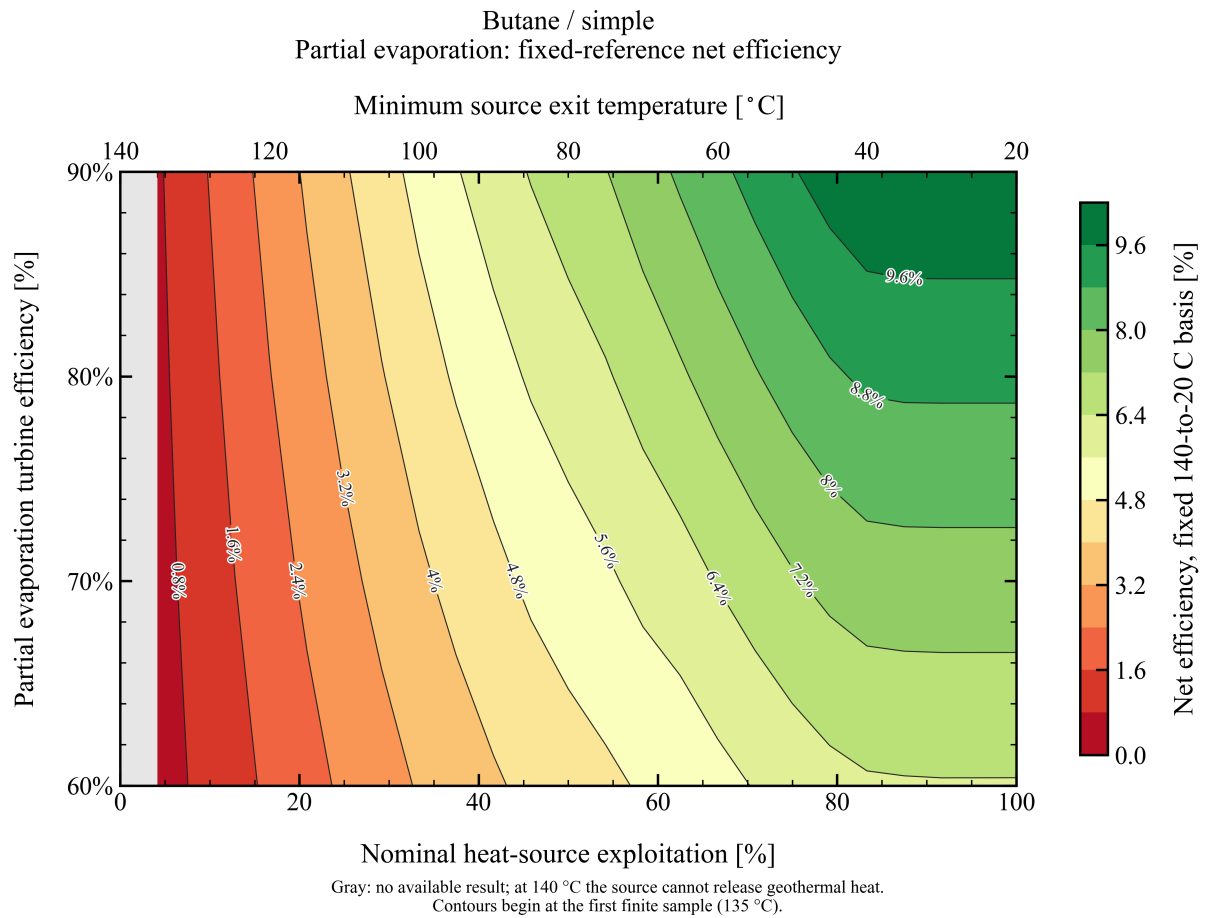
4.2 Evolution with partial evaporation



The optimized inlet has **57.00% vapor** at a 70 °C limit and **9.90%** at 20 °C. Values below 100% shorten the evaporation portion of external heating. The balance between sensible heating and evaporation can then match the cooling source differently from the superheated cycle.

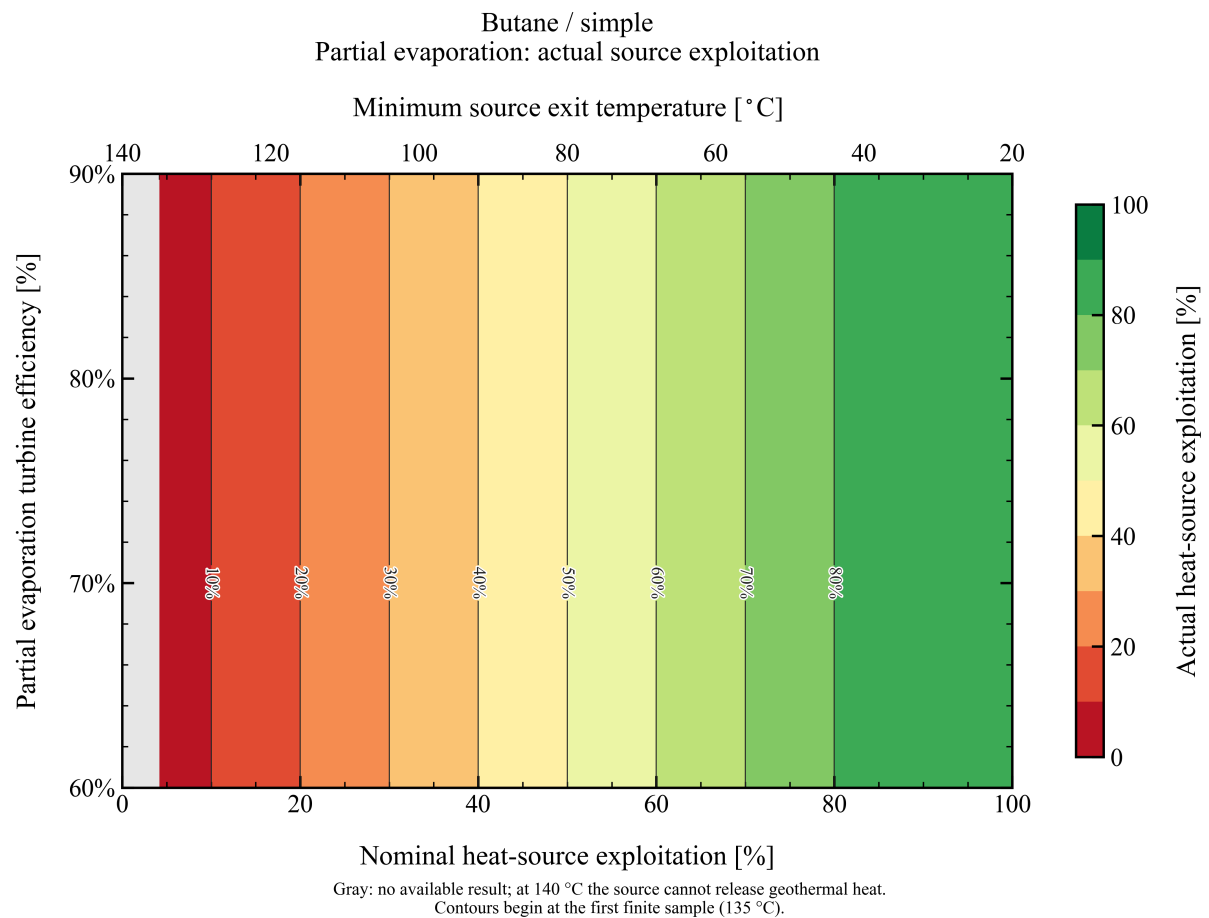
At the 70 °C limit, the partial-evaporation design admits fluid at **17.82 bar and 108.11 °C**, compared with **13.40 bar and 95.45 °C** for the baseline. Inlet-state flexibility therefore changes pressure level and temperature of heat addition, not just the amount of liquid entering the expander.

4.3 Net efficiency on the common heat basis



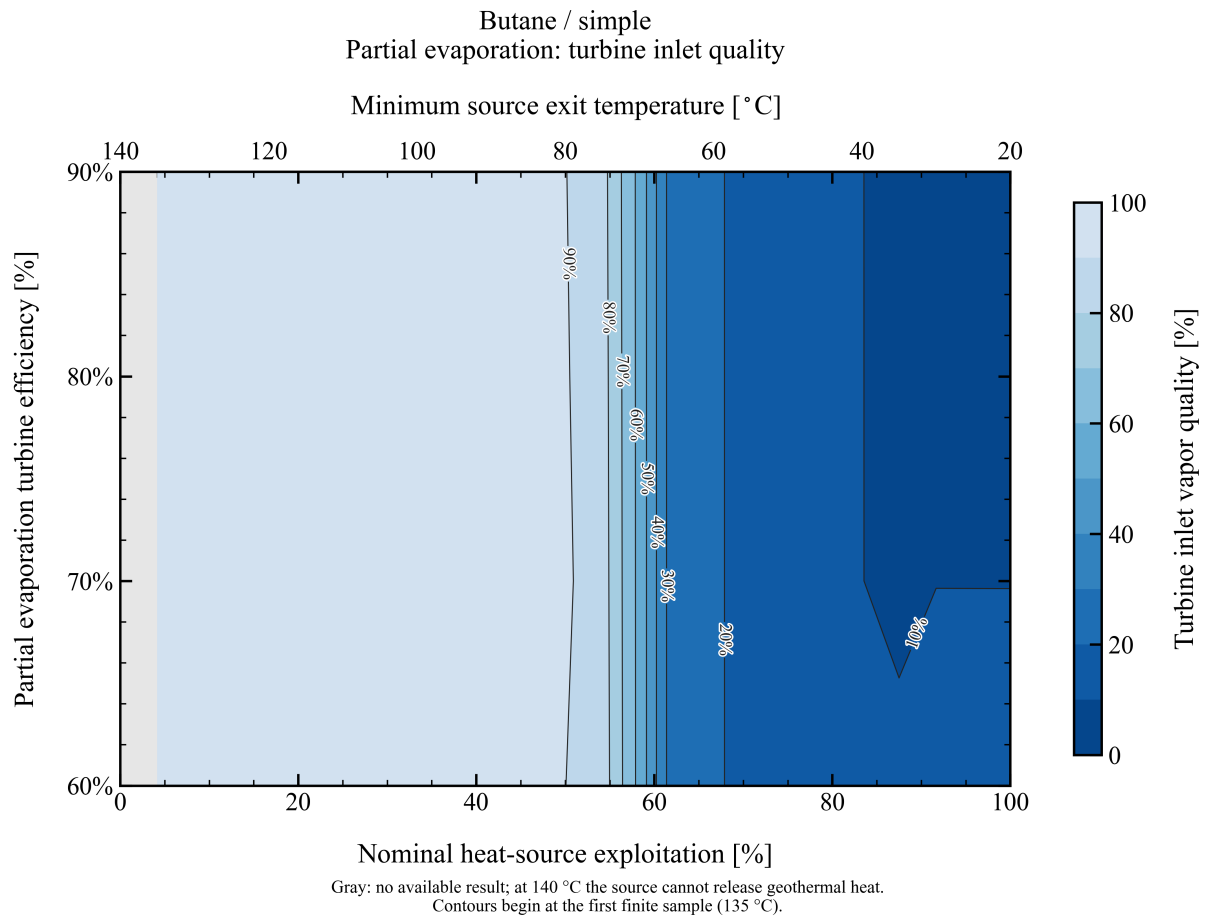
At 90% expander efficiency, the plotted efficiency is **0.69%** at a 135 °C limit, **7.65%** at 70 °C and **10.29%** at 20 °C. Unlike efficiency divided by heat available only down to each reinjection limit, this figure directly tracks net electricity per unit geothermal flow across the whole horizontal axis. Any plateau means that permitting further source cooling adds little useful power.

4.4 Actual source exploitation

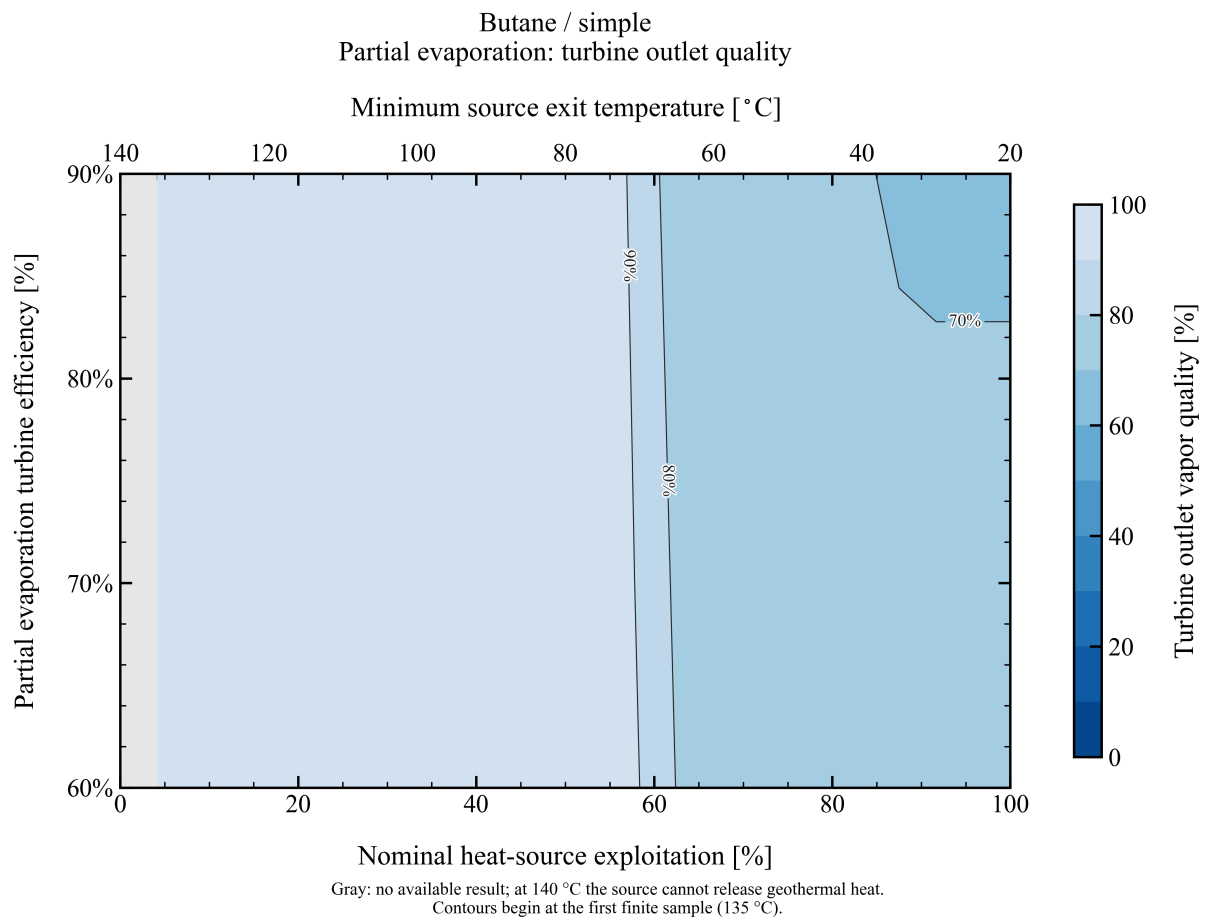


At the 20 °C limit, the four partial-evaporation designs actually leave the source at **34.66–35.78 °C**, giving **86.85–87.78%** actual exploitation. The map distinguishes allowed cooling from cooling selected by the optimizer. More recovered heat alone does not establish greater power: its temperature and the expansion efficiency also matter.

4.5 Expander inlet and outlet quality



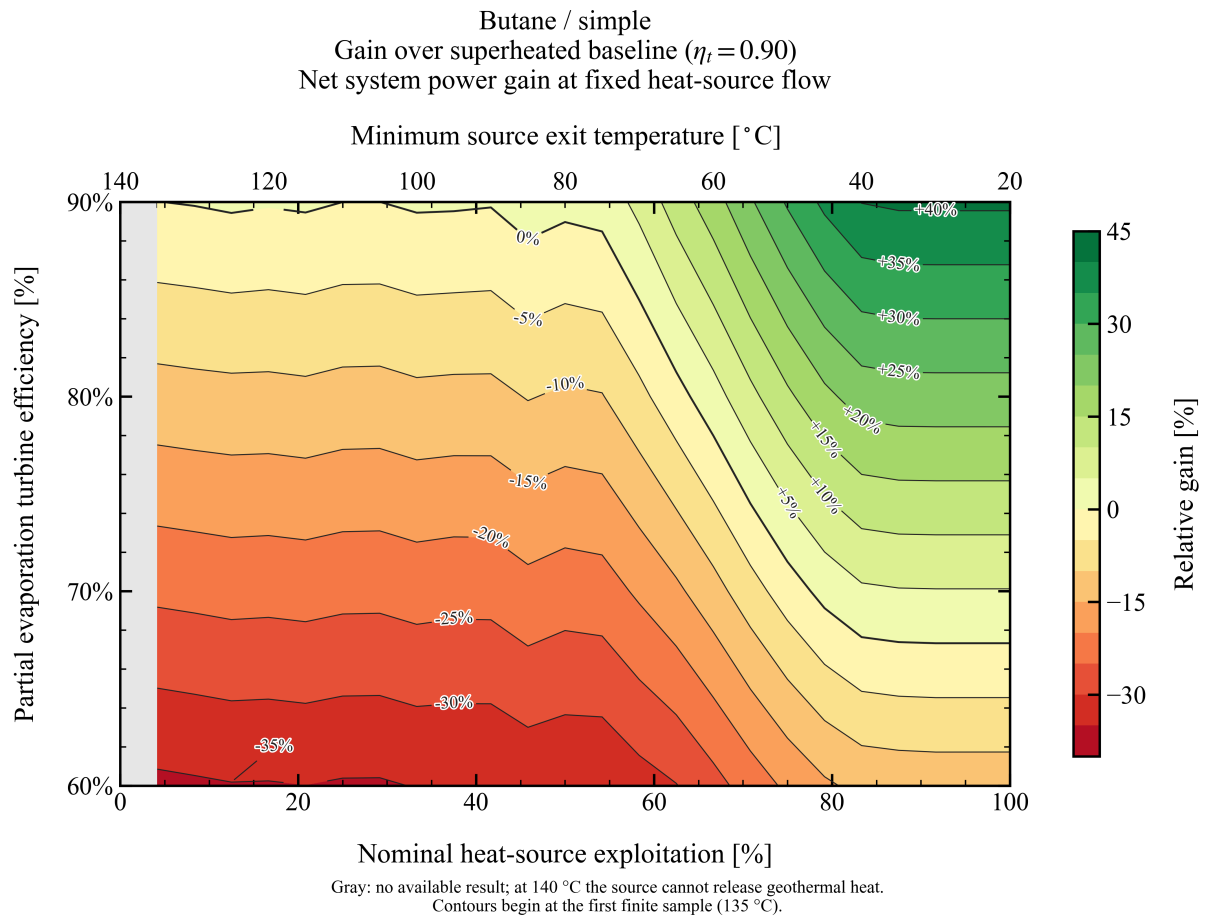
Across the 96 partial-formulation designs, **64** have two-phase inlets, **32** have dry-vapor inlets and **0** have liquid inlets. The formulation optimizes inlet state rather than imposing a wet inlet. A displayed value of 100% includes superheated vapor.



Expansion endpoint phases	Number of designs
wet to wet	45
dry to dry	32
wet to dry	19

At 90% efficiency, displayed outlet quality is **85.40%** at the 70 °C limit and **69.11%** at 20 °C. The deep-recovery design remains two-phase at discharge; its advantage therefore requires an expander suited to substantial liquid at both ends.

4.6 Relative gain and its physical origin



At the **70 $^{\circ}\text{C}$** limit and 90% expander efficiency, partial evaporation changes fixed-reference efficiency from **7.18% to 7.65%**, a **+6.45% relative gain**. The actual source outlets are **70.48 $^{\circ}\text{C}$** for the baseline and **70.00 $^{\circ}\text{C}$** for partial evaporation. The differing source outlets show that heat recovery also contributes; the gain cannot be interpreted as a change in conversion efficiency alone.

At the **20 $^{\circ}\text{C}$** limit, partial evaporation at 90% efficiency changes recovered source heat per kilogram by **+37.17%** and net system power per unit absorbed heat by **+2.66%**. These factors multiply, giving **+40.82% net-power gain** at equal geothermal flow. This separates the value of additional source cooling from the value of converting the recovered heat.

Minimum source outlet [$^{\circ}\text{C}$]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
80	50.00	+1.23%	-10.69%	-22.65%	-34.21%
70	58.33	+6.45%	-6.36%	-19.18%	-31.99%
60	66.67	+17.86%	+2.99%	-11.87%	-26.74%
50	75.00	+30.61%	+14.06%	-2.49%	-19.04%

Minimum source outlet [°C]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
40	83.33	+40.13%	+22.19%	+4.25%	-13.69%
20	100.00	+40.82%	+22.81%	+4.80%	-13.09%

4.7 Efficiency penalty and break-even conditions

At **80% expander efficiency**, the largest modeled gain is **+22.81%**. The highest sampled reinjection limit with positive gain is **60 °C**, where the gain is **+2.99%**.

At **70% expander efficiency**, the largest modeled gain is **+4.80%**. The highest sampled reinjection limit with positive gain is **45 °C**, where the gain is **+1.52%**.

At **60% expander efficiency**, the largest modeled gain is **-13.08%**. No sampled reinjection limit beats the 90%-efficient superheated baseline.

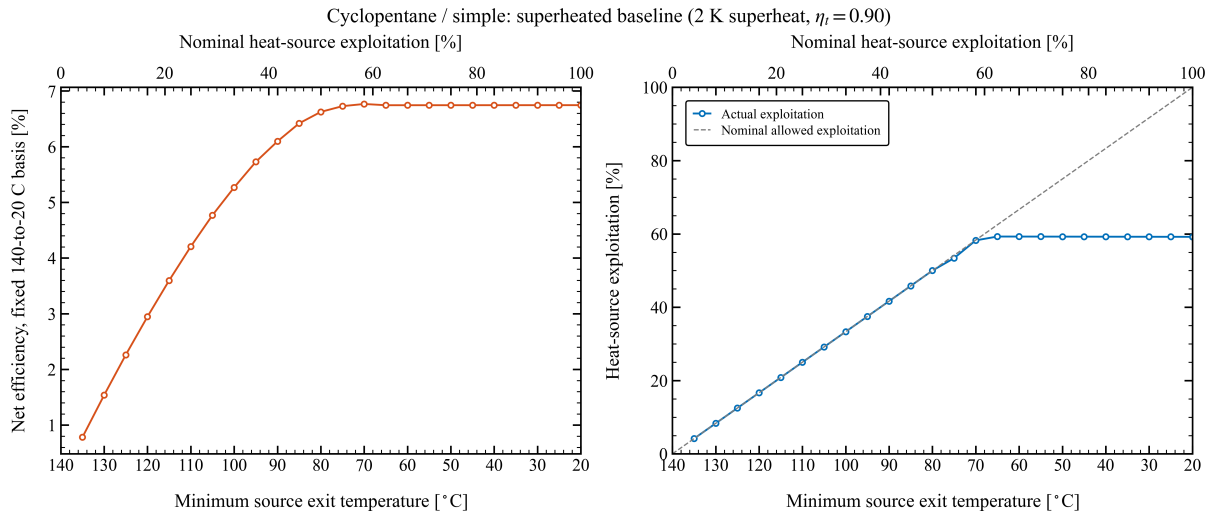
The zero contour is an approximate break-even boundary between the sampled efficiencies. A positive gain means the inlet-state and heat-recovery benefits offset the prescribed expander penalty. It does not demonstrate that a real expander can attain that efficiency at the computed inlet and outlet qualities. Near-zero gains and abrupt phase transitions are especially sensitive to local optimization and exchanger discretization.

The largest decrease in saved power per source flow relative to an earlier, stricter outlet limit is **0.01%**. Relaxing the limit preserves the earlier design's feasibility, so such a decrease is a local-solution irregularity rather than a thermodynamic penalty for allowing more cooling. It is retained under the single-start protocol.

At equal 90% expander efficiency, the partial formulation falls below the superheated baseline by more than 0.01% at 1 sampled outlet limit (largest loss **0.04%**). The flexible formulation includes the baseline's inlet state, so these negative values identify local-solution limitations, not an inherent disadvantage of allowing partial evaporation.

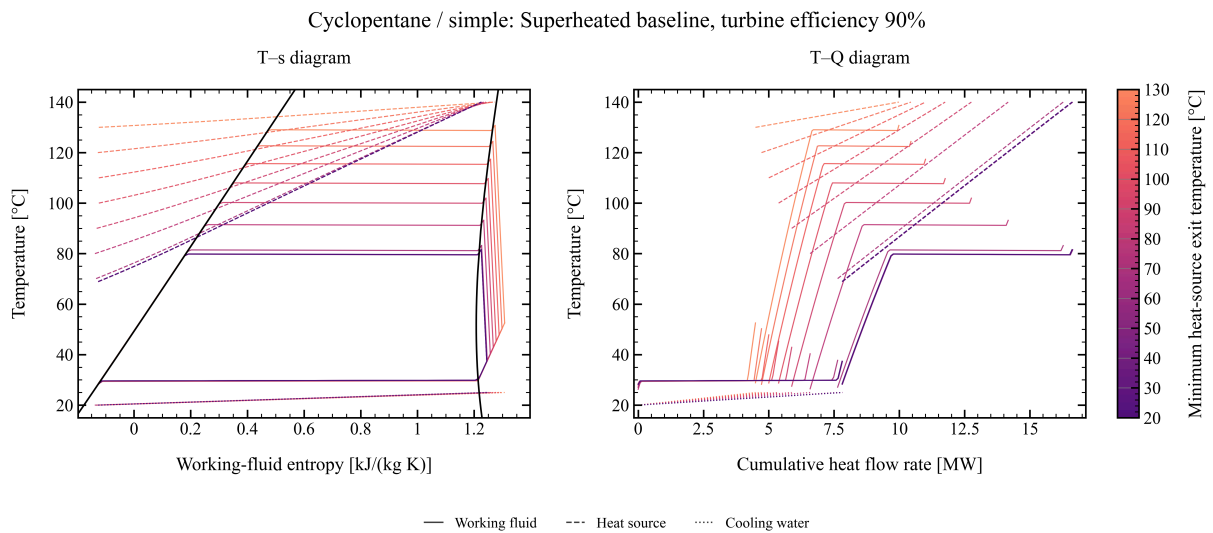
5. Cyclopentane: simple cycle

5.1 Superheated baseline



At 90% expander efficiency, the baseline's fixed-reference net efficiency changes from **0.78%** at the 135 °C reinjection limit to **6.75%** at 20 °C. Its corresponding net power per unit source flow changes from **3.95 to 34.05 kW per kg/s**.

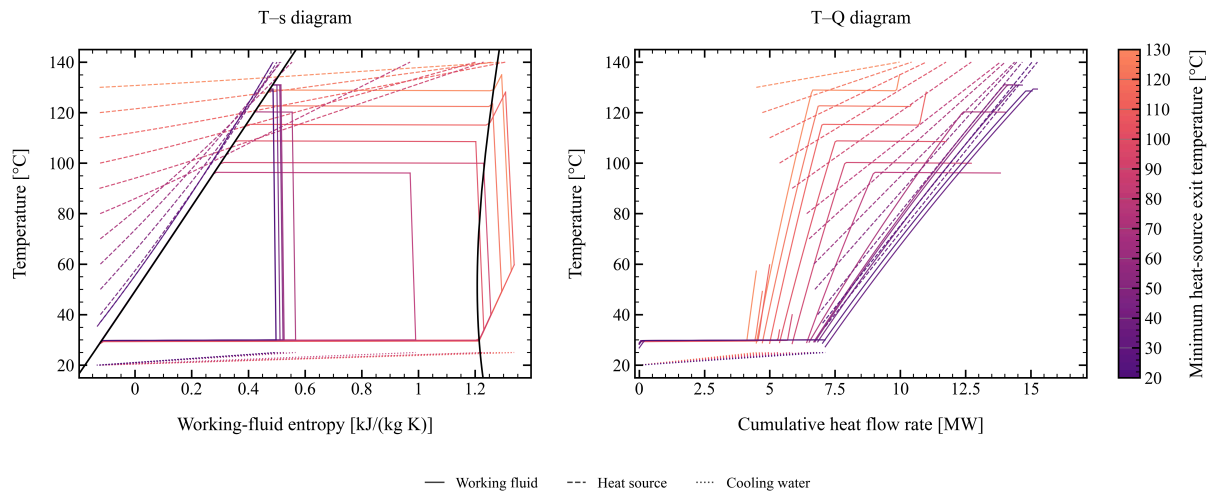
At the 20 °C limit, the baseline actually leaves the source at **68.95 °C**, or **59.21% actual exploitation**. Permission for further cooling does not require the optimizer to use it: the selected pressure levels, phase-change temperatures and exchanger approaches determine useful recovery.



The overlays show how the pressure levels and heat-addition profile change as the reinjection limit is relaxed. Coincident low-limit curves indicate that a similar design is selected despite allowing more source cooling.

5.2 Evolution with partial evaporation

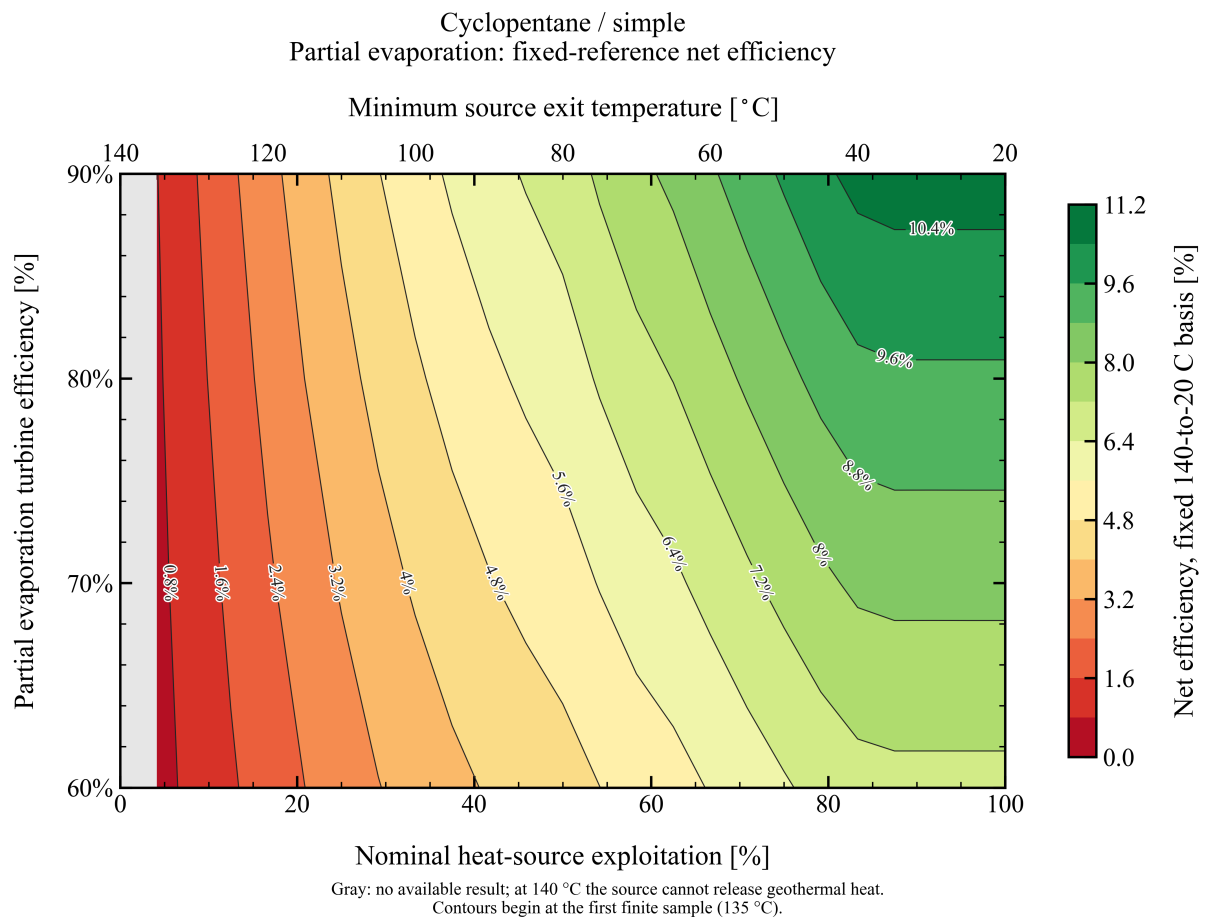
Cyclopentane / simple: Partial evaporation, turbine efficiency 90%



The optimized inlet has **15.95% vapor** at a 70 °C limit and **2.05%** at 20 °C. Values below 100% shorten the evaporation portion of external heating. The balance between sensible heating and evaporation can then match the cooling source differently from the superheated cycle. The expansion outlet must also be examined; wet admission does not imply dry discharge.

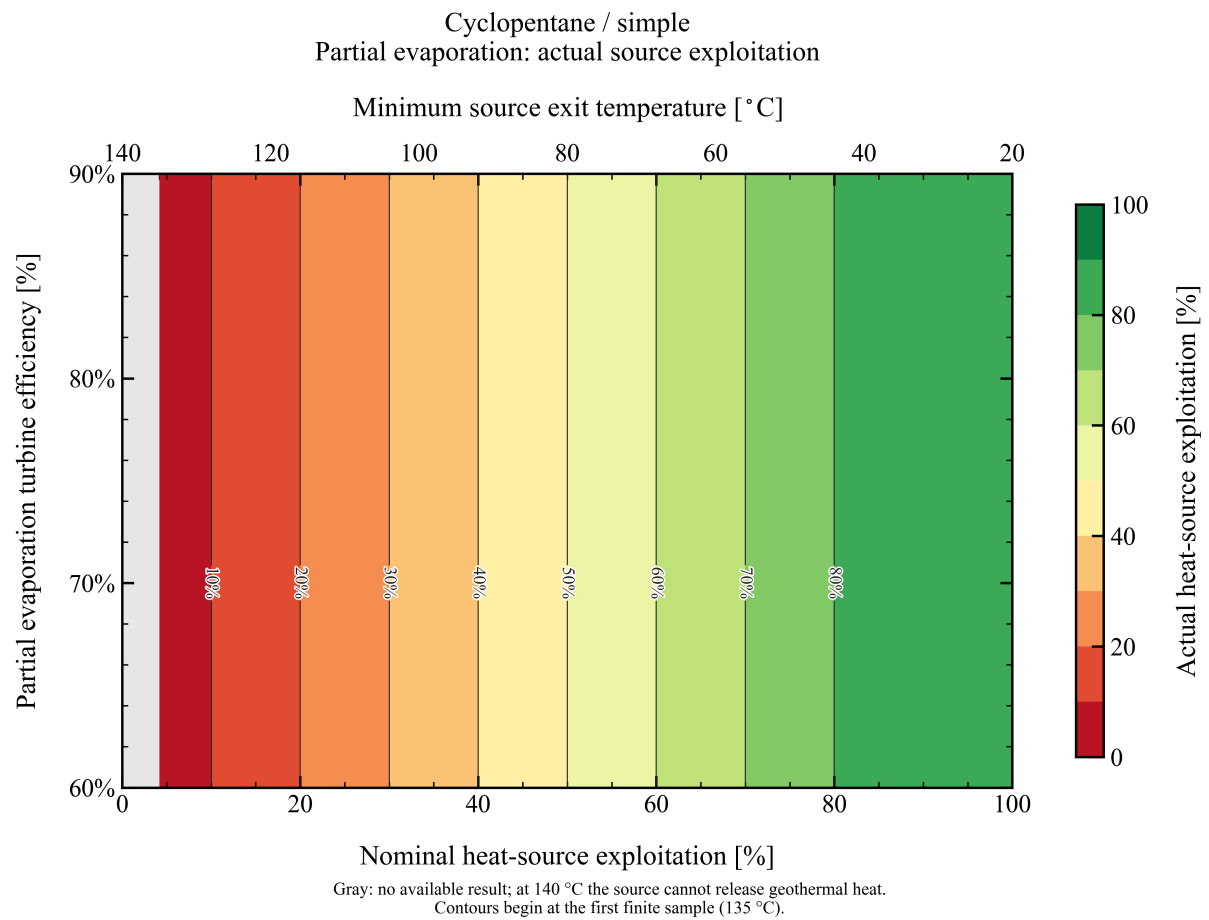
At the 70 °C limit, the partial-evaporation design admits fluid at **6.54 bar and 120.19 °C**, compared with **2.60 bar and 83.17 °C** for the baseline. Inlet-state flexibility therefore changes pressure level and temperature of heat addition, not just the amount of liquid entering the expander.

5.3 Net efficiency on the common heat basis



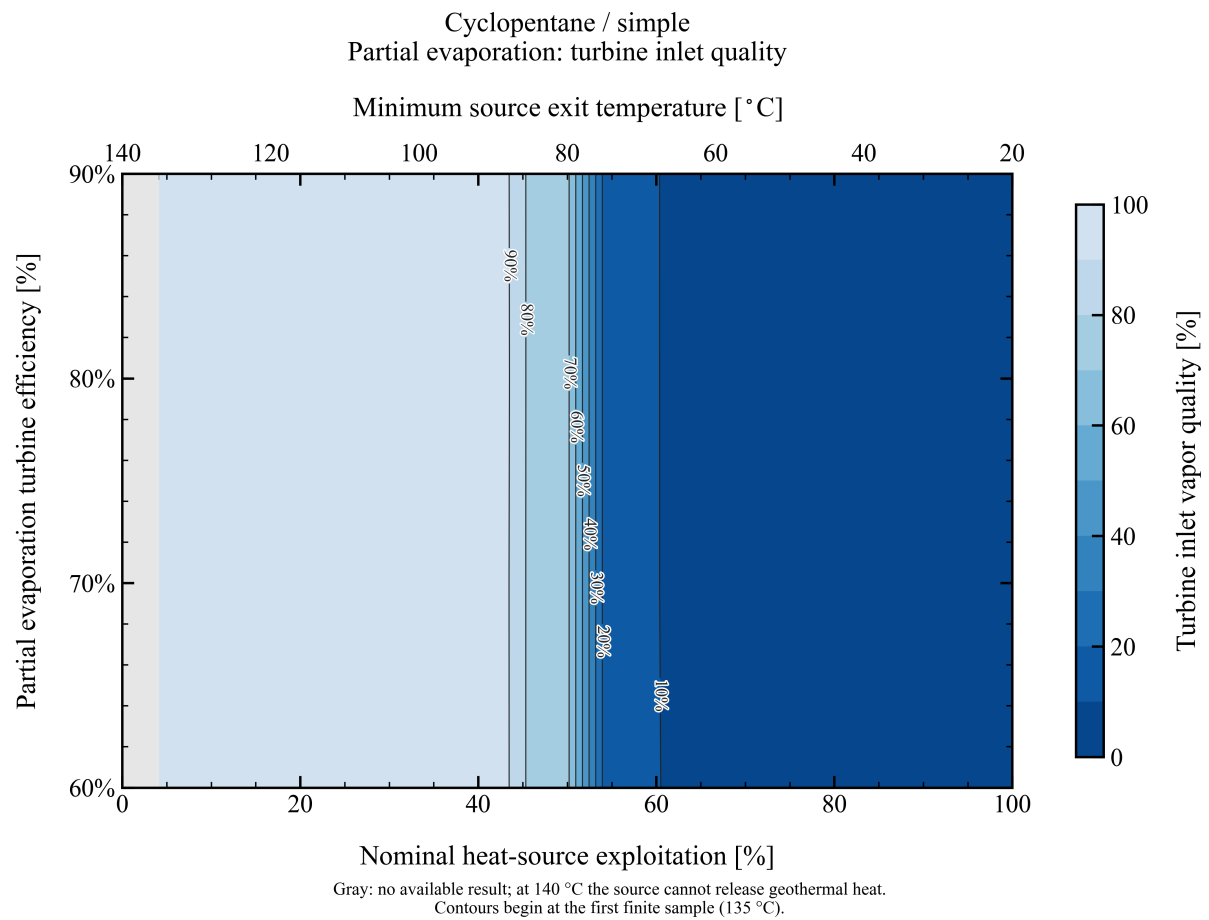
At 90% expander efficiency, the plotted efficiency is **0.78%** at a 135 °C limit, **7.80%** at 70 °C and **10.74%** at 20 °C. Unlike efficiency divided by heat available only down to each reinjection limit, this figure directly tracks net electricity per unit geothermal flow across the whole horizontal axis. Any plateau means that permitting further source cooling adds little useful power.

5.4 Actual source exploitation

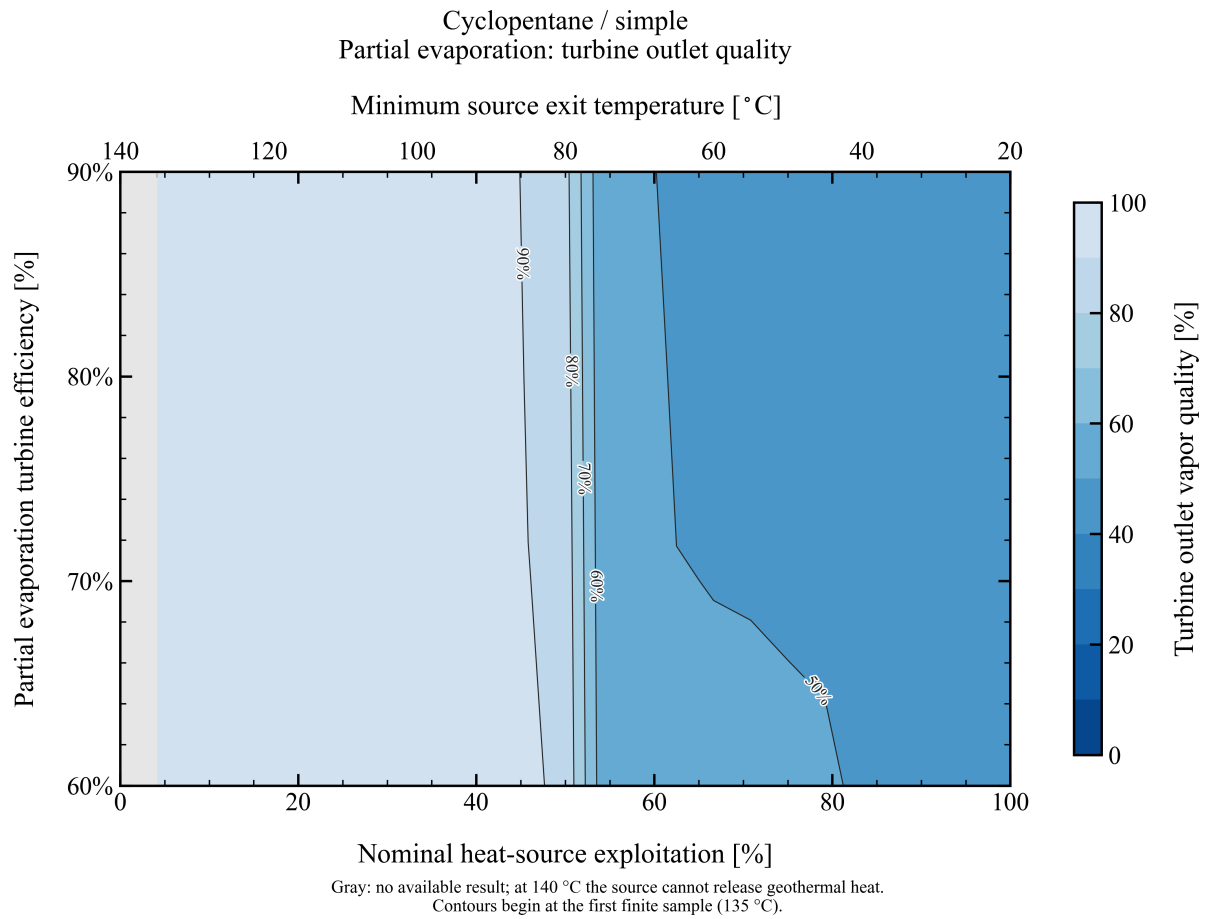


At the 20 °C limit, the four partial-evaporation designs actually leave the source at **35.46–36.62 °C**, giving **86.15–87.12%** actual exploitation. The map distinguishes allowed cooling from cooling selected by the optimizer. More recovered heat alone does not establish greater power: its temperature and the expansion efficiency also matter.

5.5 Expander inlet and outlet quality



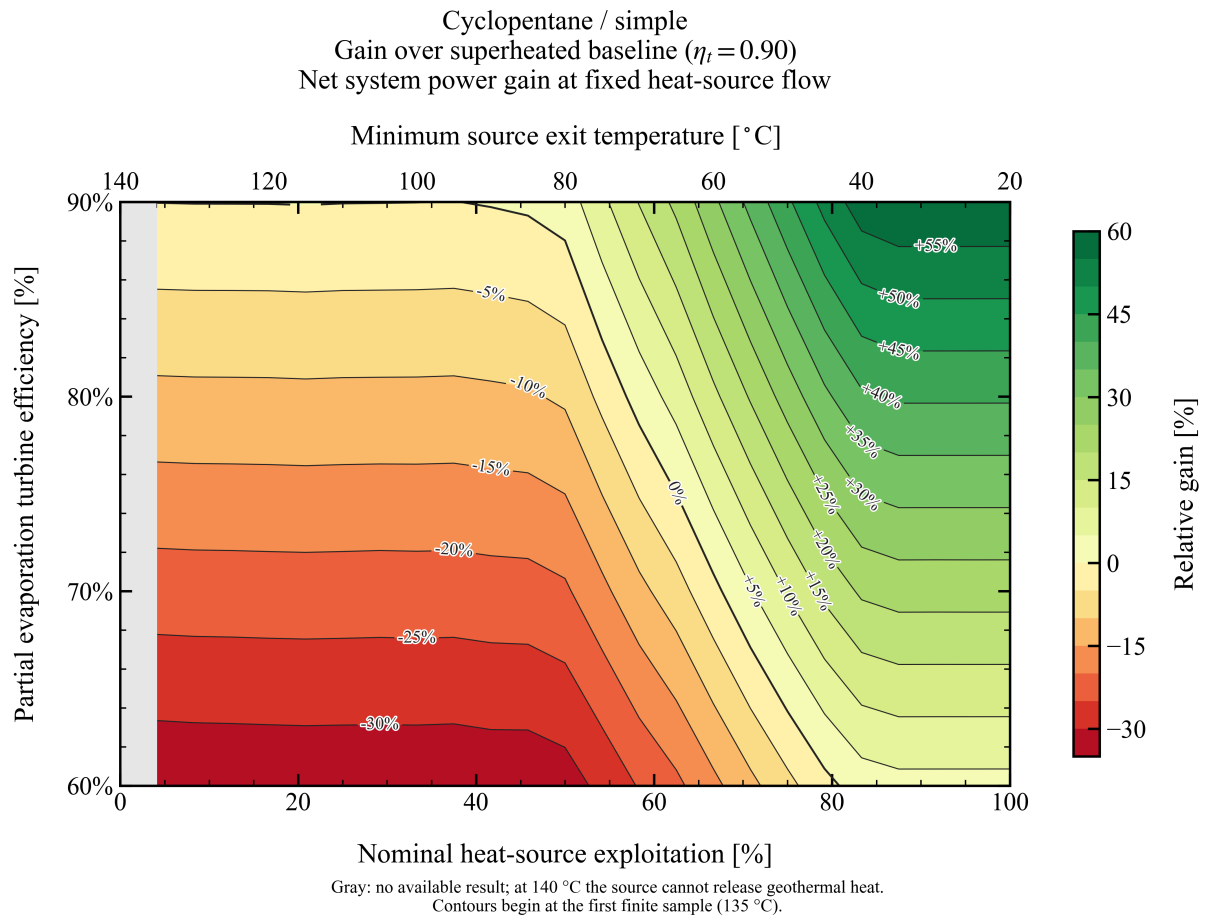
Across the 96 partial-formulation designs, **64** have two-phase inlets, **32** have dry-vapor inlets and **0** have liquid inlets. The formulation optimizes inlet state rather than imposing a wet inlet. A displayed value of 100% includes superheated vapor.



Expansion endpoint phases	Number of designs
wet to wet	56
dry to dry	32
wet to dry	8

At 90% efficiency, displayed outlet quality is **51.32%** at the 70 °C limit and **46.14%** at 20 °C. The deep-recovery design remains two-phase at discharge; its advantage therefore requires an expander suited to substantial liquid at both ends.

5.6 Relative gain and its physical origin



At the **70 $^{\circ}\text{C}$** limit and 90% expander efficiency, partial evaporation changes fixed-reference efficiency from **6.77% to 7.80%**, a **+15.24% relative gain**. The actual source outlets are **70.11 $^{\circ}\text{C}$** for the baseline and **70.00 $^{\circ}\text{C}$** for partial evaporation. The differing source outlets show that heat recovery also contributes; the gain cannot be interpreted as a change in conversion efficiency alone.

At the **20 $^{\circ}\text{C}$** limit, partial evaporation at 90% efficiency changes recovered source heat per kilogram by **+45.04%** and net system power per unit absorbed heat by **+9.81%**. These factors multiply, giving **+59.27% net-power gain** at equal geothermal flow. This separates the value of additional source cooling from the value of converting the recovered heat.

Minimum source outlet [$^{\circ}\text{C}$]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
80	50.00	+2.28%	-9.23%	-20.75%	-32.26%
70	58.33	+15.24%	+1.95%	-11.34%	-24.62%
60	66.67	+28.90%	+13.83%	-1.36%	-16.30%
50	75.00	+44.09%	+27.24%	+10.44%	-6.45%

Minimum source outlet [°C]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
40	83.33	+57.76%	+39.28%	+20.83%	+2.36%
20	100.00	+59.27%	+40.63%	+22.02%	+3.41%

5.7 Efficiency penalty and break-even conditions

At **80% expander efficiency**, the largest modeled gain is **+40.64%**. The highest sampled reinjection limit with positive gain is **70 °C**, where the gain is **+1.95%**.

At **70% expander efficiency**, the largest modeled gain is **+22.02%**. The highest sampled reinjection limit with positive gain is **55 °C**, where the gain is **+4.62%**.

At **60% expander efficiency**, the largest modeled gain is **+3.42%**. The highest sampled reinjection limit with positive gain is **40 °C**, where the gain is **+2.36%**.

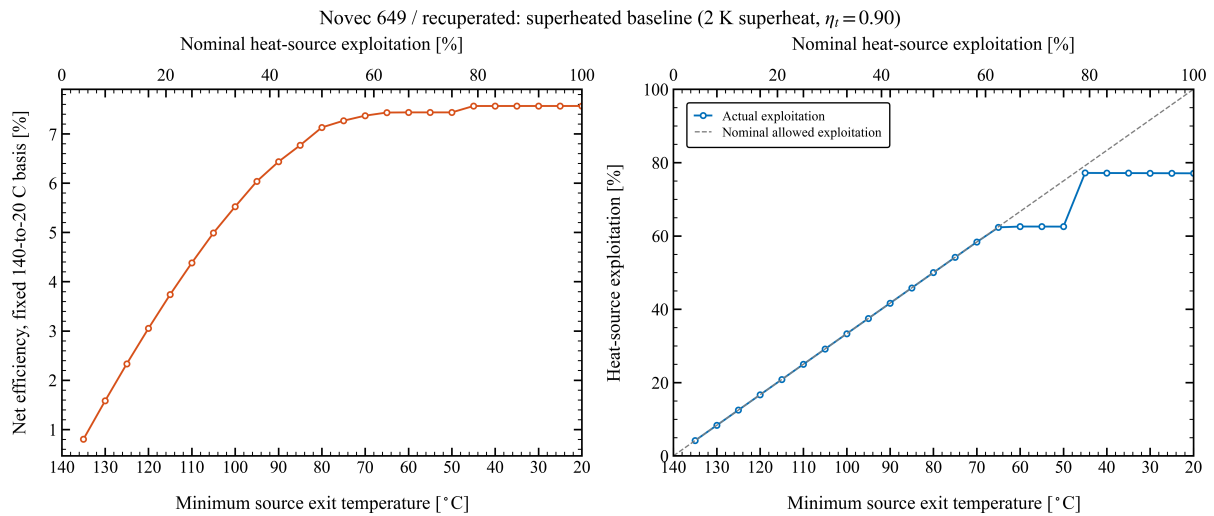
The zero contour is an approximate break-even boundary between the sampled efficiencies. A positive gain means the inlet-state and heat-recovery benefits offset the prescribed expander penalty. It does not demonstrate that a real expander can attain that efficiency at the computed inlet and outlet qualities. Near-zero gains and abrupt phase transitions are especially sensitive to local optimization and exchanger discretization.

The largest decrease in saved power per source flow relative to an earlier, stricter outlet limit is **0.33%**. Relaxing the limit preserves the earlier design's feasibility, so such a decrease is a local-solution irregularity rather than a thermodynamic penalty for allowing more cooling. It is retained under the single-start protocol.

At equal 90% expander efficiency, the partial formulation falls below the superheated baseline by more than 0.01% at 1 sampled outlet limit (largest loss **0.06%**). The flexible formulation includes the baseline's inlet state, so these negative values identify local-solution limitations, not an inherent disadvantage of allowing partial evaporation.

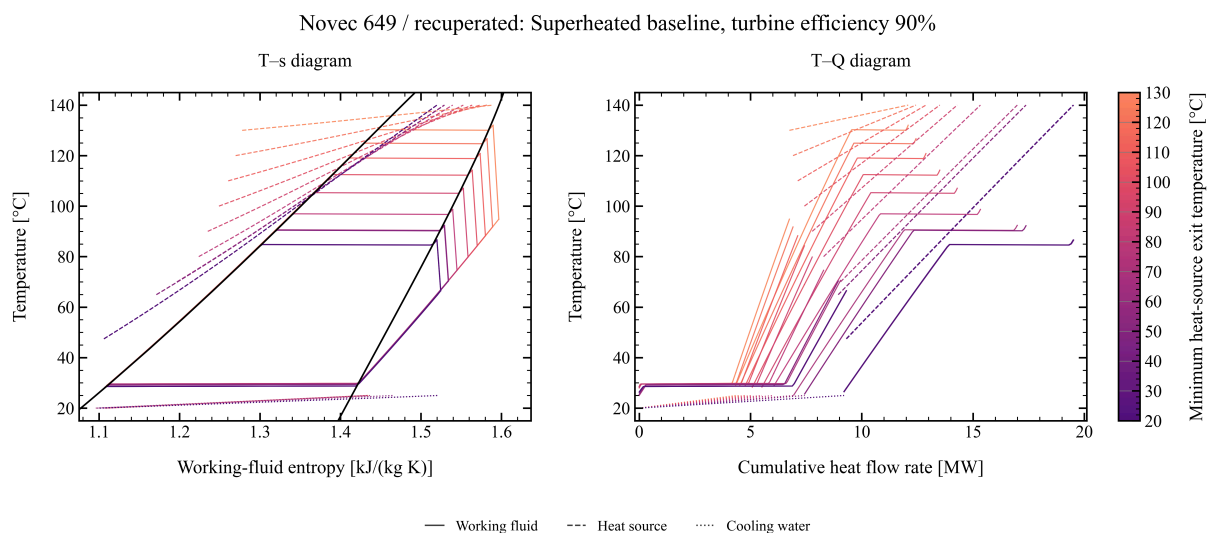
6. Novec 649: recuperated cycle

6.1 Superheated baseline



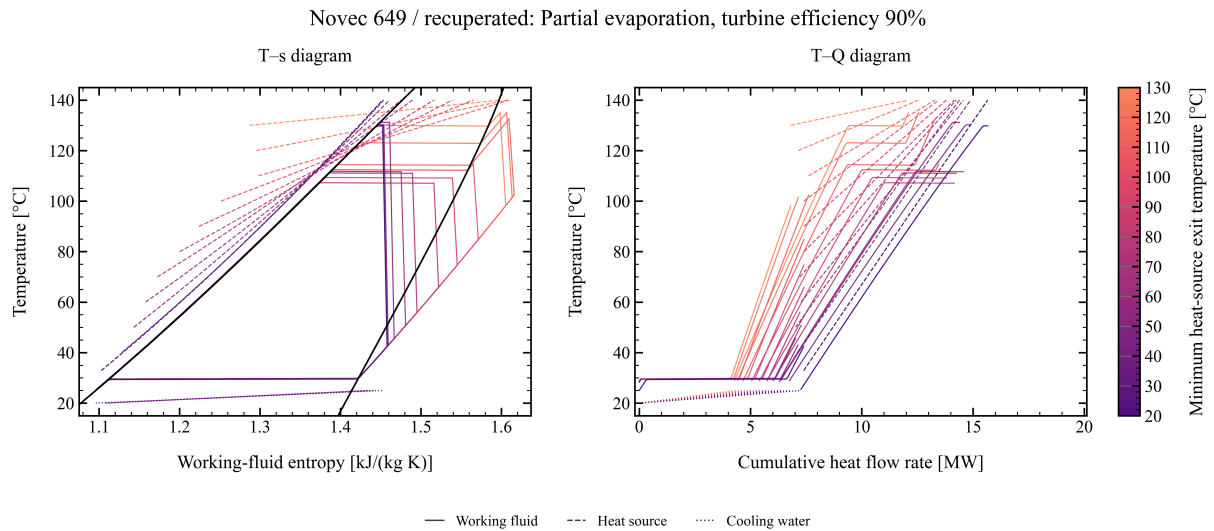
At 90% expander efficiency, the baseline's fixed-reference net efficiency changes from **0.80%** at the 135 °C reinjection limit to **7.57%** at 20 °C. Its corresponding net power per unit source flow changes from **4.05 to 38.19 kW per kg/s**.

At the 20 °C limit, the baseline actually leaves the source at **47.50 °C**, or **77.09% actual exploitation**. Permission for further cooling does not require the optimizer to use it: the selected pressure levels, phase-change temperatures and exchanger approaches determine useful recovery.



The overlays show how the pressure levels and heat-addition profile change as the reinjection limit is relaxed. Coincident low-limit curves indicate that a similar design is selected despite allowing more source cooling.

6.2 Evolution with partial evaporation

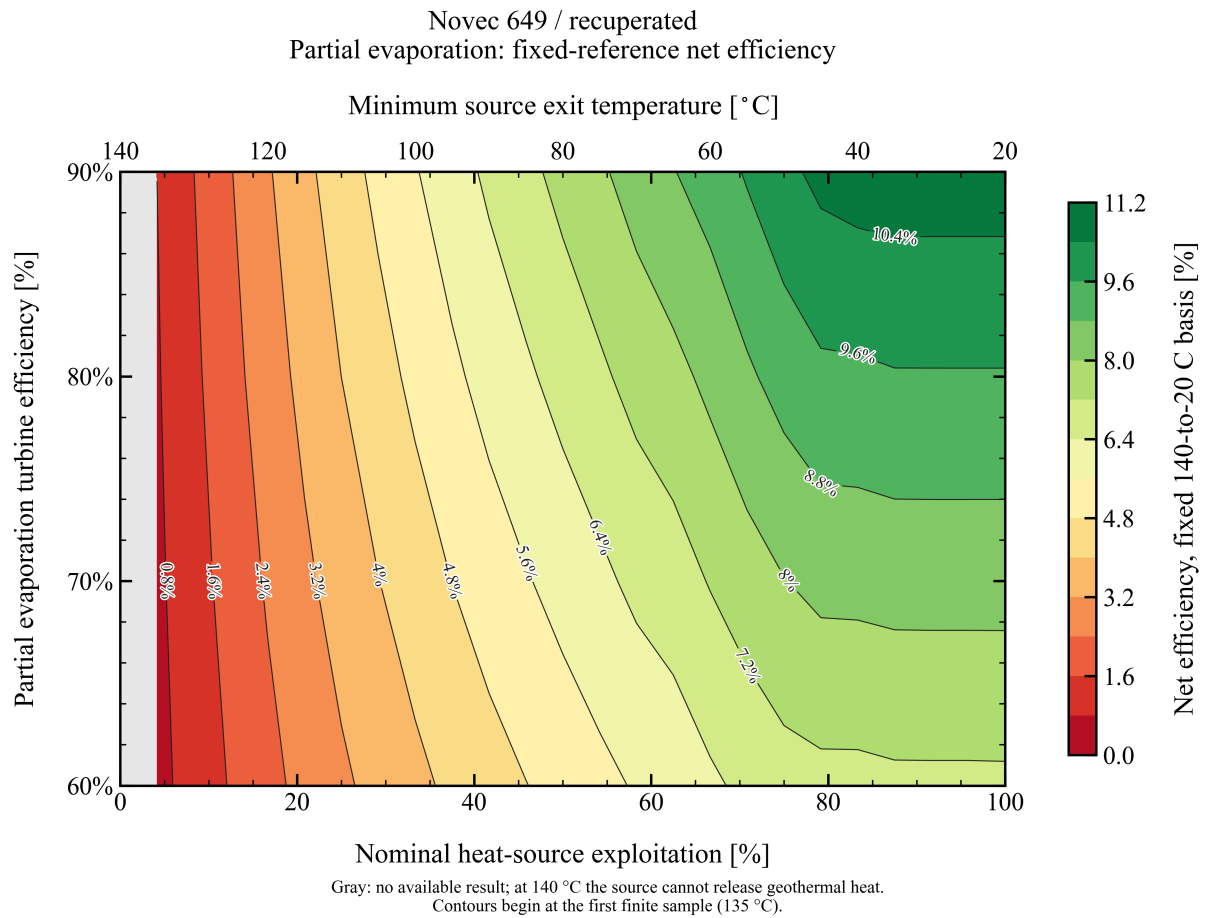


The optimized inlet has **60.80% vapor** at a 70 °C limit and **5.58%** at 20 °C. Values below 100% shorten the evaporation portion of external heating. The balance between sensible heating and evaporation can then match the cooling source differently from the superheated cycle. The expansion outlet must also be examined; wet admission does not imply dry discharge.

At the 70 °C limit, the partial-evaporation design admits fluid at **5.81 bar and 110.99 °C**, compared with **3.51 bar and 92.37 °C** for the baseline. Inlet-state flexibility therefore changes pressure level and temperature of heat addition, not just the amount of liquid entering the expander.

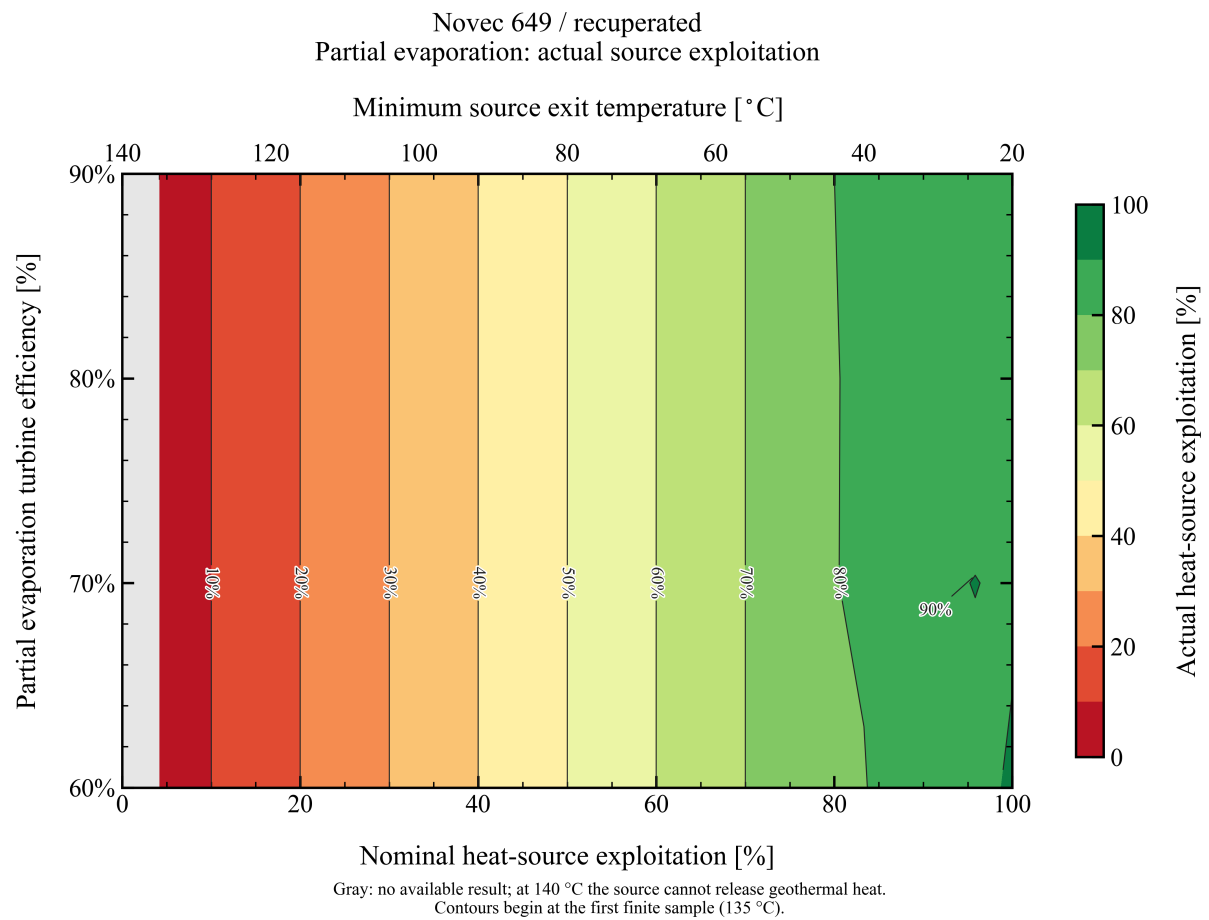
For Novec, recuperator effectiveness is **0.617/0.648** for baseline/partial evaporation at 70 °C and **0.033/0.097** at 20 °C. Internal preheating reduces external heat demand, but can also prevent deep cooling of the geothermal stream. The optimizer balances these effects; a recuperated architecture need not use high effectiveness at every condition.

6.3 Net efficiency on the common heat basis



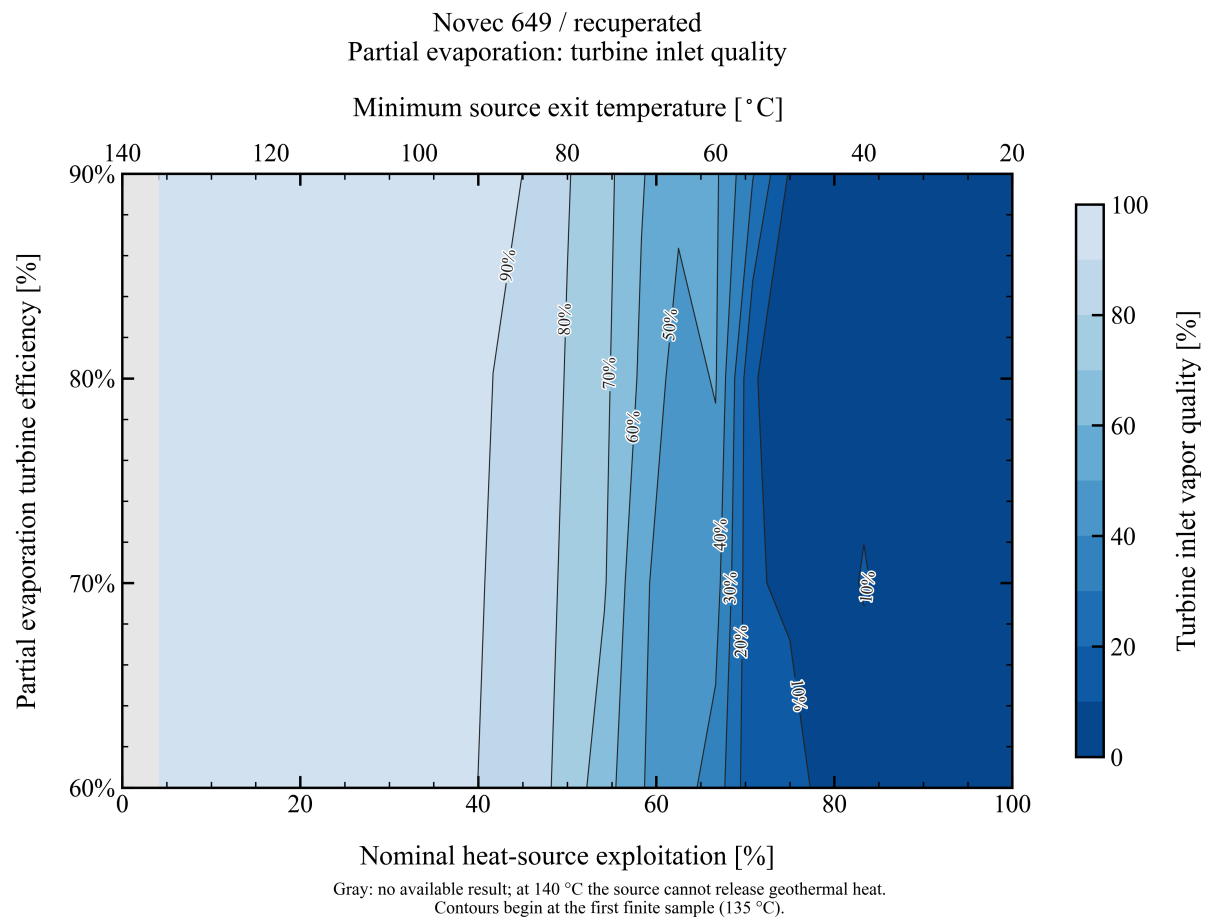
At 90% expander efficiency, the plotted efficiency is **0.80%** at a 135 °C limit, **8.34%** at 70 °C and **10.79%** at 20 °C. Unlike efficiency divided by heat available only down to each reinjection limit, this figure directly tracks net electricity per unit geothermal flow across the whole horizontal axis. Any plateau means that permitting further source cooling adds little useful power.

6.4 Actual source exploitation

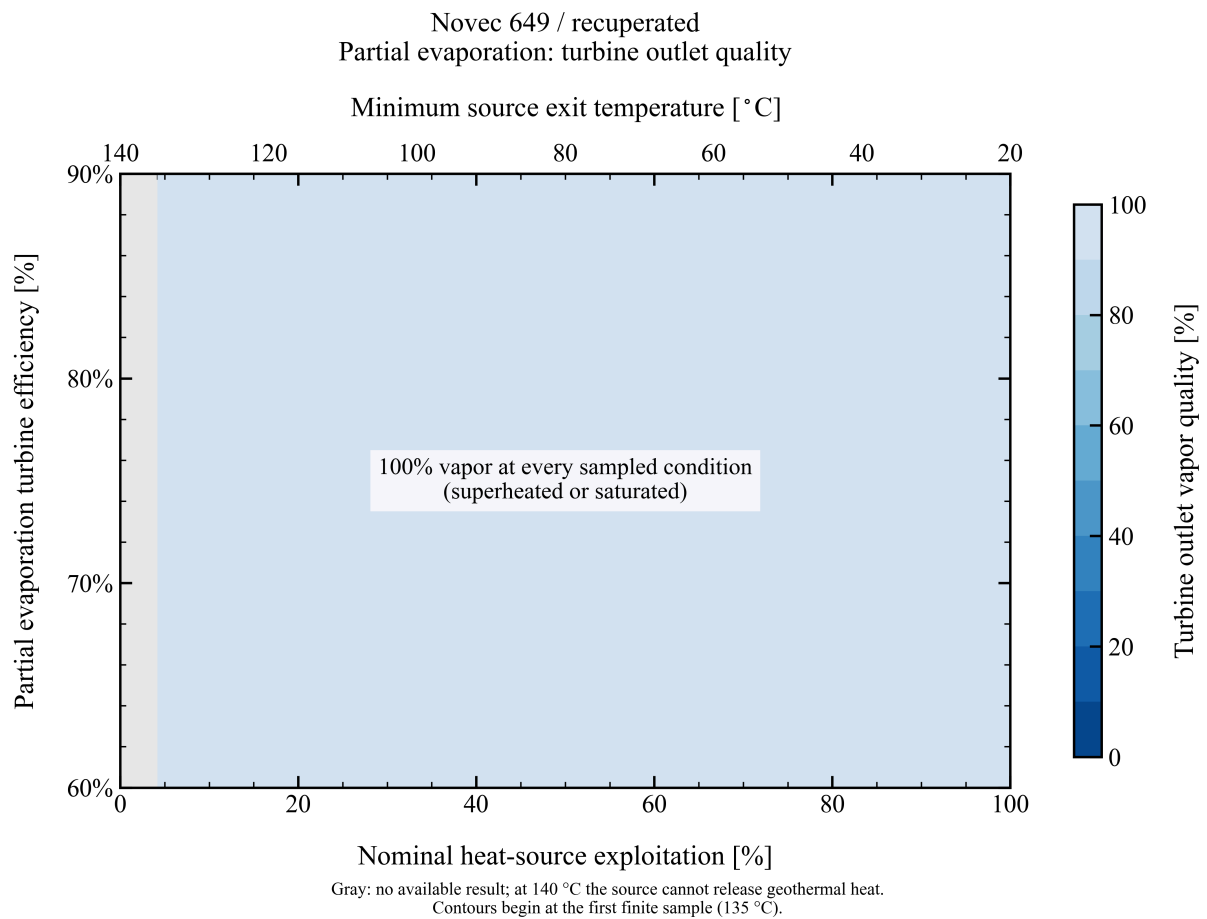


At the 20 °C limit, the four partial-evaporation designs actually leave the source at **31.54–34.04 °C**, giving **88.30–90.38%** actual exploitation. The map distinguishes allowed cooling from cooling selected by the optimizer. More recovered heat alone does not establish greater power: its temperature and the expansion efficiency also matter.

6.5 Expander inlet and outlet quality



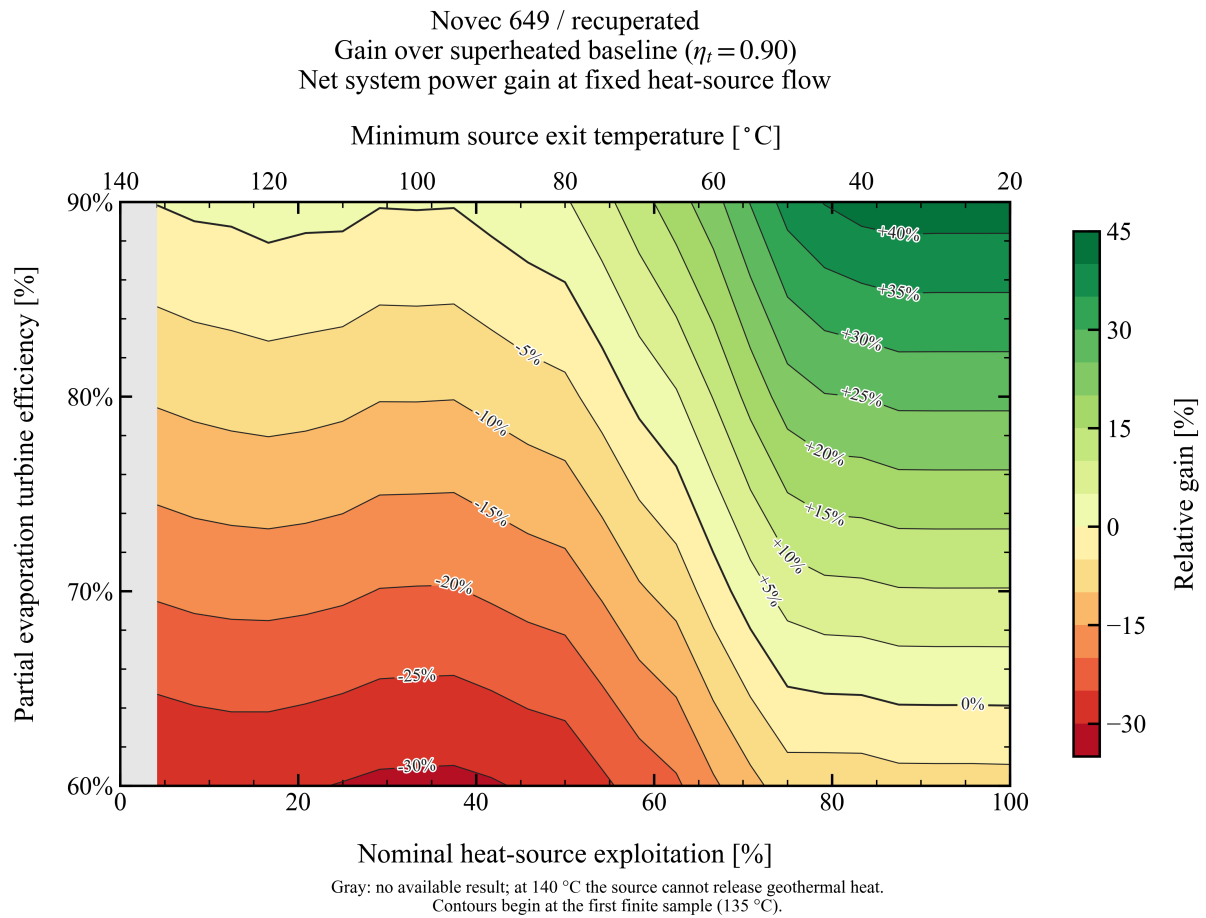
Across the 96 partial-formulation designs, **64** have two-phase inlets, **32** have dry-vapor inlets and **0** have liquid inlets. The formulation optimizes inlet state rather than imposing a wet inlet. A displayed value of 100% includes superheated vapor.



Expansion endpoint phases	Number of designs
wet to dry	64
dry to dry	32

At 90% efficiency, displayed outlet quality is **100.00%** at the 70 °C limit and **100.00%** at 20 °C. The deep-recovery design reaches dry vapor at discharge. If its inlet is wet, this is a wet-to-dry expansion; the dry outlet does not remove the need to accommodate liquid admission.

6.6 Relative gain and its physical origin



At the **70 °C** limit and 90% expander efficiency, partial evaporation changes fixed-reference efficiency from **7.37% to 8.34%**, a **+13.10% relative gain**. The actual source outlets are **70.00 °C** for the baseline and **70.00 °C** for partial evaporation. Because both reach essentially the same outlet, this gain comes from converting the recovered heat more effectively, not from a larger recovered heat quantity.

At the **20 °C** limit, partial evaporation at 90% efficiency changes recovered source heat per kilogram by **+15.66%** and net system power per unit absorbed heat by **+23.35%**. These factors multiply, giving **+42.66% net-power gain** at equal geothermal flow. This separates the value of additional source cooling from the value of converting the recovered heat.

Minimum source outlet [°C]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
80	50.00	+4.46%	-6.35%	-17.43%	-28.79%
70	58.33	+13.10%	+1.37%	-10.61%	-23.01%
60	66.67	+23.11%	+10.22%	-2.51%	-15.79%
50	75.00	+37.11%	+22.56%	+7.26%	-7.52%

Minimum source outlet [°C]	Nominal exploitation [%]	Gain at expander efficiency 90%	80%	70%	60%
40	83.33	+42.14%	+25.06%	+8.93%	-7.77%
20	100.00	+42.66%	+26.22%	+9.74%	-6.80%

6.7 Efficiency penalty and break-even conditions

At **80% expander efficiency**, the largest modeled gain is **+26.22%**. The highest sampled reinjection limit with positive gain is **70 °C**, where the gain is **+1.37%**.

At **70% expander efficiency**, the largest modeled gain is **+9.75%**. The highest sampled reinjection limit with positive gain is **55 °C**, where the gain is **+2.78%**.

At **60% expander efficiency**, the largest modeled gain is **-6.80%**. No sampled reinjection limit beats the 90%-efficient superheated baseline.

The zero contour is an approximate break-even boundary between the sampled efficiencies. A positive gain means the inlet-state and heat-recovery benefits offset the prescribed expander penalty. It does not demonstrate that a real expander can attain that efficiency at the computed inlet and outlet qualities. Near-zero gains and abrupt phase transitions are especially sensitive to local optimization and exchanger discretization.

The largest decrease in saved power per source flow relative to an earlier, stricter outlet limit is **0.07%**. Relaxing the limit preserves the earlier design's feasibility, so such a decrease is a local-solution irregularity rather than a thermodynamic penalty for allowing more cooling. It is retained under the single-start protocol.

7. Cross-fluid interpretation and limits

At a 20 °C reinjection limit and 90% expander efficiency:

Configuration	Baseline power per source flow [kW per kg/s]	Partial power per source flow [kW per kg/s]	Gain [%]	Partial fixed- reference efficiency [%]
Butane: simple cycle	36.88	51.93	+40.82	10.29
Cyclopentane: simple cycle	34.05	54.22	+59.27	10.74
Novec 649: recuperated cycle	38.19	54.48	+42.66	10.79

Relative gain and absolute electricity production answer different questions. A larger gain can reflect a weaker baseline rather than a better final cycle. The fixed-reference efficiency columns permit direct comparison of power per geothermal flow across both fluids and reinjection limits.

7.1 Remaining sampling and optimization limits

The **5 K approach is imposed at 100 evaporator/condenser locations and 25 recuperator locations, not continuously**. Nine selected designs per fluid and their coarse-grid seeds were reevaluated at 501 locations without reoptimization:

Fluid	Coarse seed: lowest checked heater approach [K]	Fine result: lowest checked heater approach [K]	Fine result: lowest checked condenser approach [K]	Fine result: lowest checked recuperator approach [K]
Butane	3.451	4.658	4.968	—
Cyclopentane	2.983	4.553	4.970	—
Novec 649	4.118	4.780	4.955	5.000

These subset diagnostics do not locate the worst approach over all designs. The reported results retain the requested 100-element evaporator and condenser model. Approaches below 5 K mean that a finer discretization and reoptimization are needed before treating precise break-even efficiencies or small gains as design conclusions. Close fluid rankings should be read with the same caution.